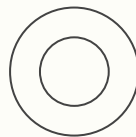


Portable CHAMPAGNE*

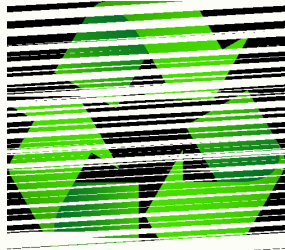
A Case Study in Rehabilitating Open
Source Repositories for Reproducibility

Team members: **Sophia K. Cheng**, Kelly Sovacool, Padmashri Saravanan

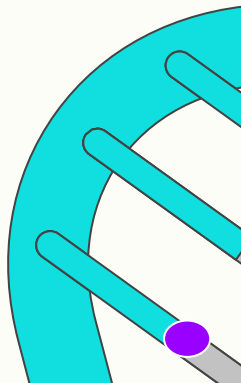
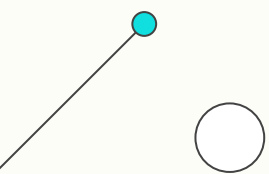
*CHAMPAGNE = **CH**romAtin **iM**muno **P**recipit**A**tion
sequencin**G** a**N**alysis pip**E**line



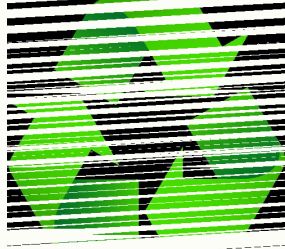
Motivation



Passion for reproducibility



Motivation

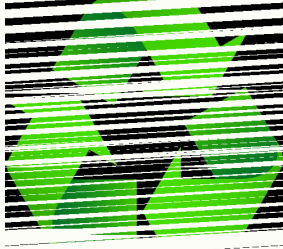


Passion for reproducibility



More than just
forking & running

Motivation



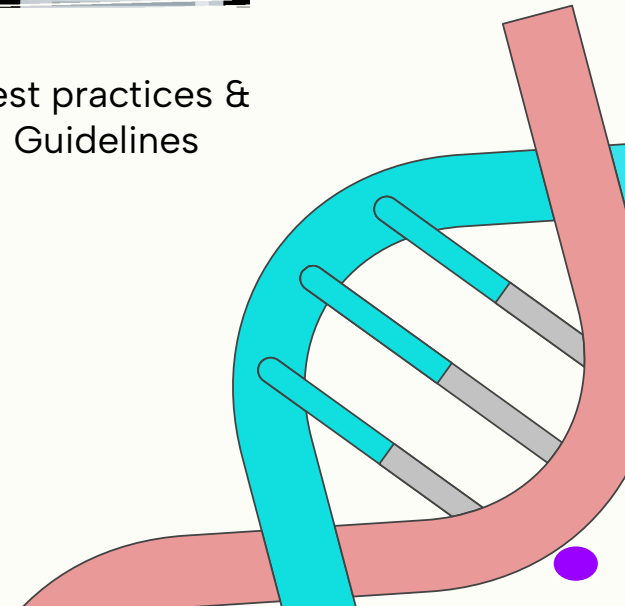
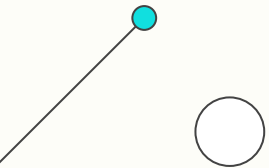
Passion for reproducibility



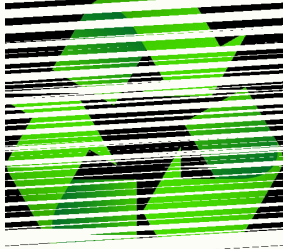
More than just
forking & running



Best practices &
Guidelines



Motivation



Passion for reproducibility

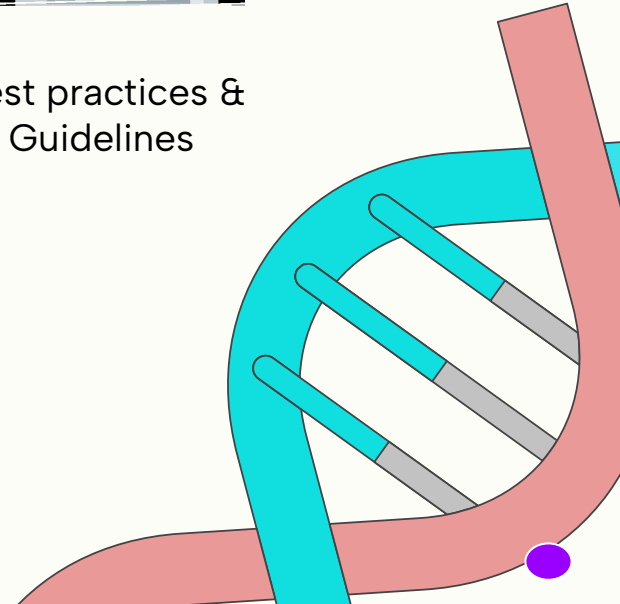


More than just
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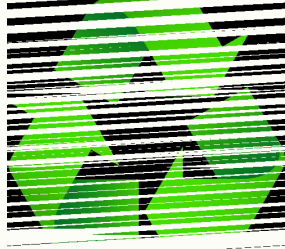


Best practices &
Guidelines

Agentic Tools
as aid



Motivation



Passion for reproducibility



More than just
forking & running

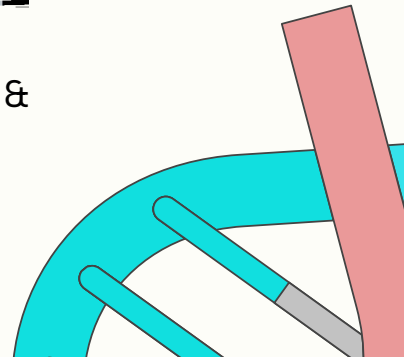


Best practices &
Guidelines

Agentic Tools
as aid



Kelly is working
on open source
workflow that
was a perfect
case study



 nextflow

Background on CHAMPAGNE

- Pipeline for analyzing sequence data from ChIP-seq experiments.
- Developed for researchers at NIH NCI on the Biowulf HPC.
- Works great on Biowulf...

Background on CHAMPAGNE

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But what about for the broader scientific community?

Project Plan

Running the
pipeline
locally



What are
dependencies we
run into outside of
Biowulf?



Project Plan

Running the
pipeline
locally

Cataloguing
Errors

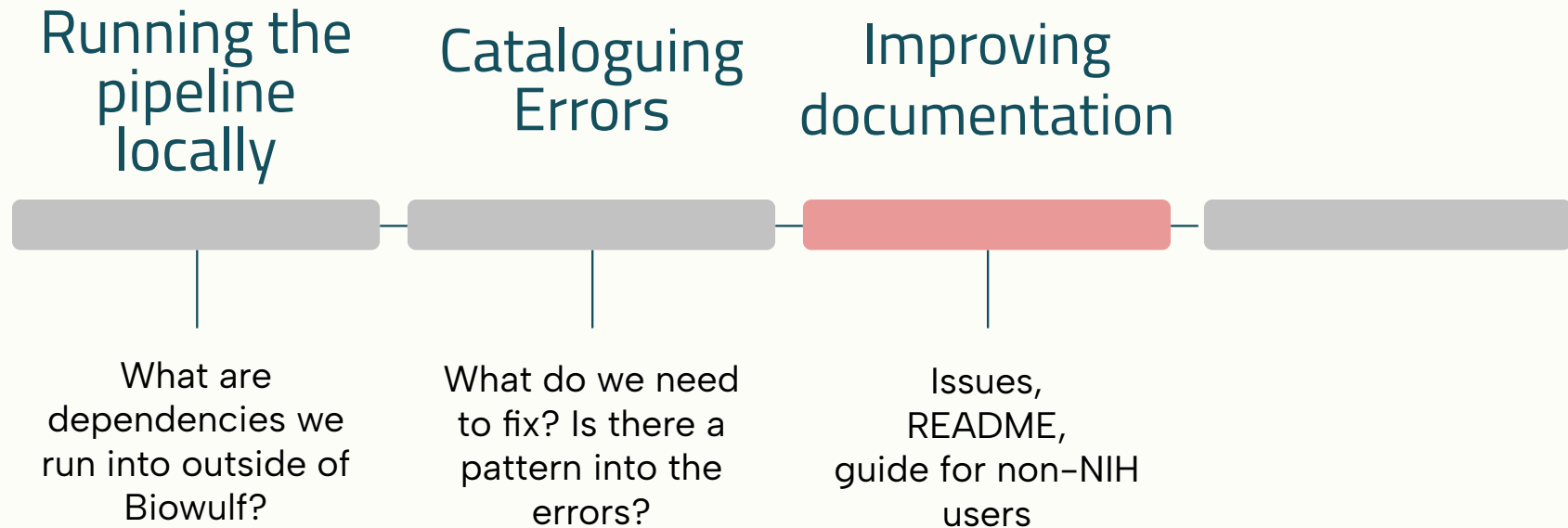


What are
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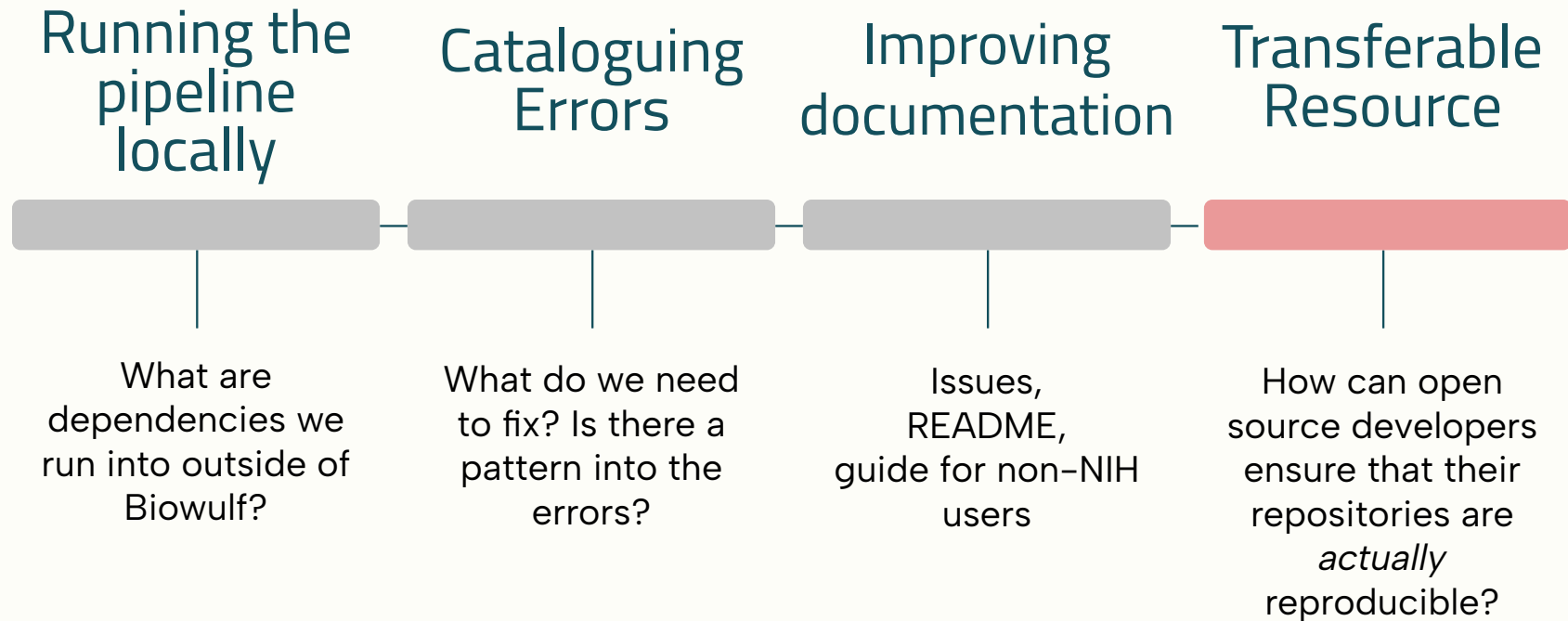
What do we need
to fix? Is there a
pattern into the
errors?



Project Plan



Project Plan

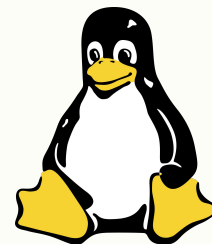
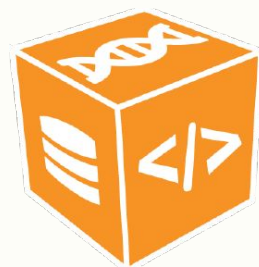
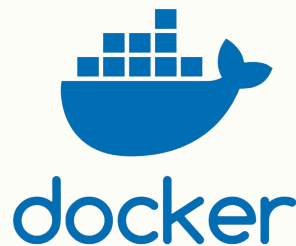


Running the pipeline locally



macOS

 Windows



LLMs that helped us

LLM	Utilization
Cursor w/PyCharm (Sophia)	Setting up & debugging for running locally on WSL2.
ChatGPT (Padma)	Debugging errors running the pipeline locally on macOS.
CoPilot in VS Code (Kelly)	Assist with minor code changes to support custom reference files, make some steps optional, and docs improvements.

Running the pipeline locally



Windows + WSL2 Ubuntu



Nextflow version 25.10.0

GNU coreutils + CLI

Latest version of conda

Apptainer downloaded

macOS

Nextflow version 25.10.0

Java, specifically version 17

Latest version of Bash

Docker downloaded



Cataloguing errors



- CHAMPAGNE required Nextflow v25.10.0 (previous release) and Java v17 (several releases ago).
- Pre-built reference genomes only on Biowulf HPC.
- FastQ Screen required Biowulf-only databases.
- Compute resources were set for HPCs, not meant for a laptop.
- Containers were not pulled from the right registry when docker was used (worked fine with singularity/apptainer).



Improvements to CHAMPAGNE

Opened 9 issues, closed 4 of them. Merged 6 pull requests.

☐	 0 Open	☒ 6 Closed	Author ▾	Label ▾
☐	 docs: specify nxf_ver=25.10.0 ✓	bosc-cofest-2026	#356 by kelly-sovacool (Member) was merged 1 hour ago • Review required	3 tasks done
☐	 docs: fix broken link to diagram ✓	bosc-cofest-2026	#354 by kelly-sovacool (Member) was merged 2 hours ago • Review required	3 tasks done
☐	 ci: set nxf v25.10.0 in ci ✓	bosc-cofest-2026	#353 by kelly-sovacool (Member) was merged 2 hours ago • Review required	4 tasks
☐	 fix: make docker container spec compatible with docker engine ✓	bosc-cofest-2026	#352 by kelly-sovacool (Member) was merged 1 hour ago • Review required	2 tasks done 2026-07
☐	 fix: skip fastq screen if conf not set ✓	bosc-cofest-2026	#350 by kelly-sovacool (Member) was merged 1 hour ago • Review required	3 tasks done
☐	 fix: skip blacklist filtering if no blacklist is available ✓	bosc-cofest-2026	#349 by kelly-sovacool (Member) was merged 1 hour ago • Review required	3 tasks done 2026-07

Portability-first checklist



- ❑ Dependencies are defined in containers.
- ❑ Default compute resources can be customized by the user.
- ❑ Required files are provided, or there is automation to rebuild them.
E.g. support standard pre-built reference genomes, and also allow users to bring their own custom genomes.
- ❑ No hard-coded absolute paths. Paths are relative to the repo or working directory.
- ❑ Provide a small test configuration + dataset for new users to run the software.





Now to turn this into a resource for the open source community....



GitHub Action!

You can now add this workflow as an action to any of your bioinformatics repositories!



bit.ly/repo-to-go



GitHub Action!

bit.ly/repo-to-go

A screenshot of a web browser displaying the GitHub repository page for 'caffindFrog / repo-to-go'. The browser's address bar shows 'github.com/caffindFrog/repo-to-go/tree/main'. The repository page includes a header with navigation tabs (Code, Issues, Pull requests, Agents, Actions, Projects, Wiki, Security and quality, Insights, Settings) and repository statistics (Watch: 0, Fork: 0, Starred: 2). Below the header is a table of files and folders, each with a commit message and timestamp. The 'README' file is selected, showing the repository's description and a small version of the 'repo-to-go' logo. The right sidebar contains sections for 'About', 'Releases', 'Packages', and 'Contributors'.

File/Folder	Commit Message	Time Ago
.idea	initial draft	1 hour ago
examples	fix: syntax	15 minutes ago
open-checklist	fix: syntax	15 minutes ago
resources	adding the logo	21 minutes ago
LICENSE	Initial commit	1 hour ago
README.md	chore: Merge branch 'main' of https://github.com/caffind...	15 minutes ago
agent-portability-guide.md	initial draft	1 hour ago
checklist.md	docs: link to repo	10 minutes ago

repo-to-go
Automating a checklist for making bioinformatics repositories portable and reproducible.
Portability Checklist

Future Considerations



For CHAMPAGNE:

- How do other pipelines handle FastQ Screen outside of Biowulf?
Feature: add ability to build or retrieve DBs.

For repo-to-go:

- Adding in-depth examples to the checklist and turning the repo into a guide.
- More repositories yet to be audited which will bring more reproducibility challenges to streamline.



Thank you!

bit.ly/repo-to-go

